HE2003 Econometrics I
Tutorial 9

28/10/2024

Question 1

Explain the following terms or results; Use specific example, formula, or graph if necessary.

(a) Given an empirical example of joint null and alternative hypotheses.

(b) Explain the restricted and unrestricted regression models.

(c) Explain the formula and properties of the homoskedasticity-only F-test statistic.

(d) Show that for the OLS regression with multiple regressors, $R^2 = 0$ if $\hat{\beta}_1 = 0,\ \ \hat{\beta}_2 = 0, \dots,$ and $\hat{\beta}_k = 0$

Question 1 (a)

As an example, for the empirical application of parents' education and baby's birthweight:

$$bwght_i = \beta_0 + \beta_1 motheduc_i + \beta_2 fatheduc_i + \beta_3 cig_i + \beta_4 parity_i + \beta_5 faminc_i + u_i, \ \ i = 1, \dots, n$$

the following joint null hypothesis imposes two restrictions ($q = 2$):

$$H_0\text{: } \beta_1 + \beta_2 = 0.1, and \ \beta_3 = 0$$

which means the combined effect of parents' education is equal to 0.1 (pound/year of education), and the effect of smoking is zero. Its alternative hypothesis is

$$H_1 \text{ : at least one restriction in } H_0 \text{ does not hold}$$

This alternative hypothesis is equivalent to (when q = 2 there are 3 possible cases):

$$H_1\text{: } \beta_1 + \beta_2 \neq 0.1, or \ \beta_3 \neq 0, \text{ or neither restriction holds}$$

For reference (no need to recite): In the above example, there are 3 possible cases under H_1.
In general, if there are q restrictions in H_0, there will be $2^q - 1$ different cases under H_1.

Question 1 (b)

For example, consider the following regression model:

$$sleep_i = \beta_0 + \beta_1 totalwork_i + \beta_2 educ_i + \beta_3 age_i + u_i, \qquad i = 1, \ldots, n$$

This is the unrestricted regression model or the full regression model, and we can compute R_u^2 from this model.

Suppose we want to test the joint null hypothesis H_0: $\beta 2 = \beta 3 = 0$, so that $q = 2$ in this case (the effects of education and age are zero). Then, the restricted regression model is the model with H_0 imposed:

$$sleep_i = \beta_0 + \beta_1 totalwork_i + u_i, \qquad i = 1, \ldots, n$$

from which we compute R_r^2.

Question 1 (c)

The formula of the homoskedasticity-only F-statistic is as follows:

$$F = \frac{(R_u^2 - R_r^2)/q}{(1 - R_u^2)/(n - k - 1)}$$

where R_u^2 is the R^2 from the unrestricted regression (the full regression model), R_r^2 is the R^2 from the restricted regression under the restrictions in H_0, q is the number of restrictions in H_0, k is the number of regressors in the unrestricted regression (intercept not included in k), and n is the number of observations.

The properties of the homoskedasticity-only F-statistic is as follows:

By construction, $F \geq 0$, because R^2 increases with the number of regressors, so that R_u^2 cannot be smaller than R_r^2. The homoskedasticity-only F-statistic is comparing the fittings of the two regressions; if the “unrestricted” model fits the data sufficiently better than the “restricted” model, this can be understood as an evidence against the joint null hypothesis H_0. In this case, we will reject the “restricted” model (that is, we will reject H_0).

Question 1 (d)

Given that $\sum_{i=1}^{n} \hat{u}_i = 0 \Rightarrow \frac{1}{n}\sum_{i=1}^{n} \hat{u}_i = 0$, we have:

$$\frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \cdots - \hat{\beta}_k X_{ki}\right) = 0 \Rightarrow \bar{Y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{X}_1 - \cdots - \hat{\beta}_k \bar{X}_k = 0$$

If $\hat{\beta}_1 = 0, \ldots, \ \hat{\beta}_k = 0$, then $\bar{Y} = \hat{\beta}_0$.

On the other hand, by the definition of the OLS predicted value is

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \cdots + \hat{\beta}_k X_{ki} = \hat{\beta}_0$$

Finally, given the definition of ESS, we have $ESS = \sum_{i=1}^{n}\left(\hat{Y}_i - \bar{Y}\right)^2 = \sum_{i=1}^{n}\left(\hat{\beta}_0 - \hat{\beta}_0\right)^2 = 0$,

which implies $R^2 = ESS/TSS = 0$.

Question 2

To test the effectiveness of a job training program on the subsequent wage of workers, we specify the model:

$$wage_i = \beta_0 + \beta_1 train_i + \beta_2 educ_i + \beta_3 experience_i + u_i, \qquad i = 1, \dots, n$$

where $train_i$ is a binary regressor equal to 1 if the i-th person participated in the training program. Think of the error term as containing unobserved worker ability. Discuss in what kind of situation the OLS estimator of β_1 is likely to have bias and discuss the sign of the bias. In what kind of situation there will be no bias for the OLS estimator of β_1?

[Hint: This depends on how the employer assign the workers to the job training program (e.g., assigned randomly or not).]

Question 2

We use the OVB formula to discuss the likely bias in the OLS estimator of β_1. Because the error term contains workers' innate ability, we can specify the true model as: $(= u_i)$

$$wage_i = \beta_0 + \beta_1 train_i + \beta_2 educ_i + \beta_3 experience_i + \beta_4 ability_i + e_i, \qquad i = 1, \dots, n$$

where "$ability$" is the omitted variable. By the meaning of multiple regression, β_4 is the effect of workers' innate ability (cognitive, physical, etc.) on wage (after controlling for the other factors "$train$", "$educ$", and "$experience$").

The formula for the OVB:

$$OVB = \beta_4 \frac{Cov(train_i, ability_i)}{Var(train_i)}$$

β_4 is positive because we expect that on average the wage increases when a worker's innate ability is relatively high. Now we need to decide the sign of $Cov(train_i, ability_i)$, which will also be the sign of OVB.

Question 2

1) Suppose that the employer decided to send the workers with relatively bad performance to training, then less able workers have a greater chance of being selected for the training program, so that $Cov(train_i, ability_i) < 0$ and we have $OVB < 0$. That is, a downward bias in $\hat{\beta}_1$ in this scenario.

2) Suppose that the employer decided to send the workers with relatively good performance to training, then more able workers have a greater chance of being selected for the training program. Therefore, in this case, $Cov(train_i, ability_i) > 0$ and $OVB > 0$ (upward bias).

3) Suppose that the employer decided to randomly assign the workers to training, then we have $Cov(train_i, ability_i) = 0$ (because $train_i$ and $ability_i$ are independent in this case) and $OVB = 0$ (no bias).

Question 3

Data were collected from a random sample of 200 home sales from a community in 2013. Let $Price$ denote the selling price (in $1000s), BDR denote the number of bedrooms, $Bath$ denote the number of bathrooms, $Hsize$ denote the size of the house (in square feet), $Lsize$ denote the lot size (in square feet), Age denote the age of the house (in years), and $Poor$ denote a binary variable that is equal to 1 if the condition of the house is reported as “poor”. An estimated regression yields:

$$\widehat{Price} = \underset{(22.1)}{109.7} + \underset{(1.23)}{0.567}BDR + \underset{(9.76)}{26.9}Bath + \underset{(0.021)}{0.239}Hsize + \underset{(0.00072)}{0.005}Lsize + \underset{(0.23)}{0.1}Age - \underset{(12.23)}{56.9}Poor$$

Question 3

$$\widehat{Price} = 109.7 + 0.567BDR + 26.9Bath + 0.239Hsize + 0.005Lsize + 0.1Age - 56.9Poor$$

a) What is the predicted loss in value if a homeowner lets his house run down so that its condition becomes “poor”?

$$\Delta\, \widehat{Price} = -56.9 \times \$1{,}000 = -\$56{,}900$$

b) Suppose that a homeowner adds a new bathroom to her house, which increases the size of the house by 80 square feet. What is the predicted increase in the value of the house?

$$\Delta\, \widehat{Price} = (26.9 \times 1 + 0.239 \times 80) \times \$1{,}000 = 46.02 \times \$1{,}000 = \$46{,}020$$

Question 3

c) Is the coefficient on BDR statistically significantly different from zero at the 5% level? Explain. Construct the 95% confidence interval for the coefficient on BDR.

For testing $H0$: $\beta_1 = 0$, we have:

$$|t| = \left|\frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}\right| = \left|\frac{0.567}{1.23}\right| = 0.461 < 1.96$$

Therefore, the coefficient on BDR is not statistically significantly different from zero at the 5% level as H_0 is not rejected.

The 95% confidence interval for the coefficient on BDR is:

$$\left[\hat{\beta}_1 - 1{,}96 \times SE(\hat{\beta}_1), \hat{\beta}_1 + 1{,}96 \times SE(\hat{\beta}_1)\right]$$
$$= [0.567 - 1.96 \times 1.23, 0.567 + 1.96 \times 1.23]$$
$$= [-1.8438, 2.9778]$$

Question 3

d) Suppose you would like to test the hypothesis that BDR and Age do not matter for the selling price. Write down the joint null and alternative hypotheses. Explain the restricted and unrestricted regressions in this case.

For $i = 1, \dots, n$, the unrestricted model regression can be written as:

$$Price_i = \beta_0 + \beta_1 BDR_i + \beta_2 Bath_i + \beta_3 Hsize_i + \beta_4 Lsize_i + \beta_5 Age_i + \beta_6 Poor_i + u_i$$

The joint null hypothesis to be tested is

$$H_0: \beta_1 = 0, and\ \beta_5 = 0$$

The alternative hypothesis is

$$H_1: \beta_1 \neq 0, or\ \beta_5 \neq 0, or\ both \neq 0$$

The restricted regression model can be written as:

$$Price_i = \beta_0 + \beta_2 Bath_i + \beta_3 Hsize_i + \beta_4 Lsize_i + \beta_6 Poor_i + u_i$$

Question 4

The following equation estimates the effect of computer usage on wages.

$$wage_i = \beta_0 + \beta_1 compwork_i + \beta_2 comphome_i + \beta_3 compwork_i \times comphome_i + u_i \,, i = 1, \ldots, n$$

where u_i denotes the error term. The dummy variable $compwork$ equals one if the individual uses a computer at work and zero otherwise. The dummy variable $comphome$ equals one if the individual uses a computer at home and zero otherwise. Explain the meaning of $\beta_0, \beta_1, \beta_2$, and β_3.

[Hint: compare all the four different combinations of the values of the dummy variables $compwork$ and $comphome$.]

Question 4

To simplify notation, for w, $h = 0{,}1$, let us denote

$$wage_{[compwork=w,comphome=h]} = E[wage_i | compwork_i = w, comphome_i = h]$$

Then, given that OLS Assumption 1 holds, we have

$$\begin{aligned}
wage_{[compwork=0,comphome=0]} &= \beta_0 \\
wage_{[compwork=1,comphome=0]} &= \beta_0 + \beta_1 \\
wage_{[compwork=0,comphome=1]} &= \beta_0 + \beta_2 \\
wage_{[compwork=1,comphome=1]} &= \beta_0 + \beta_1 + \beta_2 + \beta_3
\end{aligned}$$

Comparing the first and second equations, we obtain

$$\beta_1 = wage_{[compwork=1,comphome=0]} - wage_{[compwork=0,comphome=0]}$$

Therefore, β_1 is equal to the difference in average wages between those who only use computer at work and those who never use computer.

Question 4

Comparing the first and third equations, we obtain

$$\beta_2 = wage_{[compwork=0, comphome=1]} - wage_{[compwork=0, comphome=0]}$$

Therefore, β_2 is equal to the difference in average wages between those who only use computer at home and those who never use computer.

Comparing the first and fourth equations, we obtain:

$$\beta_1 + \beta_2 + \beta_3 = wage_{[compwork=1, comphome=1]} - wage_{[compwork=0, comphome=0]}$$

Therefore, $\beta_1 + \beta_2 + \beta_3$ is equal to the difference in average wages between those who use computer both at work and at home and those who never use computer. Note that this effect is not simply adding β_1 and β_2. Here, β_3 is the **extra effect** (also called the **interaction effect**) of using computer both at work and at home (for example, one might perform more efficiently by using computer at both places and thus earn more wages/bonus. Then, β_3 will be positive in this case).

Question 4

<u>Note</u>:

Both $compwork$ and $comphome$ are dummy variables taking values 1 and 0, so there is **no slope parameter** in this model, since slope parameter is only for continuous variables such as STR and GPA. In fact, here we have specified **four different levels of wages** for four types of people: those who never use computer, those who only use at work, those who only use at home, and those who use at both places. The model in this question is called a (binary variable**) interaction regression model**, which is very useful in practice when we want to study the interaction effect of certain variables.

Question 5

Empirical Exercise: Use the dataset **CPS08.dta** uploaded in NTULearn.

(a) Run a regression of average hourly earnings (ahe) on age (age). What is the estimated effect of age on earnings?

(b) Run a regression of ahe on age, gender ($female$, equal to 1 if female, 0 otherwise), and education ($bachelor$, equal to 1 if university degree, 0 otherwise). Suppose that β_1 is the effect of age on earnings (after controlling for gender and education). Can we reject $H_0: \beta_1 = 0$ at the 5% significance level? Explain your conclusion using the values of t-test, p-value, and confidence interval reported in the software.

(c) Bob is a 26-year-old male worker with a high school diploma. Predict Bob's earnings using the estimated regression in (b). Jane is a 30-year-old female worker with a university degree. Predict Jane's earnings using this regression.

Question 5

```
. summarize
```

Variable	Obs	Mean	Std. Dev.	Min	Max
ahe	7,711	18.97609	10.13944	2.003205	82.41758
year	7,711	2008	0	2008	2008
bachelor	7,711	.4810012	.4996713	0	1
female	7,711	.4326287	.4954724	0	1
age	7,711	29.57723	2.855258	25	34

```
. table bachelor female
```

	female	
bachelor	0	1
0	2,537	1,465
1	1,838	1,871

Question 5 (a)

```
. reg ahe age, robust

Linear regression                               Number of obs     =      7,711
                                                F(1, 7709)        =     225.70
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0290
                                                Root MSE          =     9.9919
```

ahe	Coef.	Robust Std. Err.	t	P>|t|	[95% Conf.	Interval]
age	.6049863	.0402701	15.02	0.000	.526046	.6839266
_cons	1.082275	1.167013	0.93	0.354	-1.205388	3.369938

the estimated effect of age on earnings is about 0.60

Question 5 (b)

```
. reg ahe age female bachelor, robust

Linear regression                               Number of obs     =      7,711
                                                F(3, 7707)        =     555.48
                                                Prob > F          =     0.0000
                                                R-squared         =     0.1998
                                                Root MSE          =     9.0718
```

ahe	Coef.	Robust Std. Err.	t	P>|t|	[95% Conf.	Interval]
age	.5852144	.0365302	16.02	0.000	.5136052	.6568236
female	-3.664026	.2076129	-17.65	0.000	-4.071003	-3.257048
bachelor	8.083001	.2126945	38.00	0.000	7.666062	8.49994
_cons	-.6356977	1.083075	-0.59	0.557	-2.758818	1.487423

Question 5 (b)

With the multiple regression, the estimated effect of age on earning is 0.58. For testing $H_0: \beta_1 = 0$, we can reject at the 5% level because:

- $t = 16.02 > 1.96$
- $p\text{–value} = 0.000 < 0.05 = 5\%$
- The 95% confidence interval does not include 0

We also note that R^2 increased from 0.0298 in the single regression to 0.1998, but the estimate for the effect of age remains very similar (0.60 vs 0.58). This can be explained by looking at $corr(age, female)$ and $corr(age, bachelor)$. Given the small correlations, controlling for female and bachelor or not will not largely affect the estimate for age

```
. cor age female bachelor
(obs=7,711)
```

	age	female	bachelor
age	1.0000		
female	-0.0307	1.0000	
bachelor	0.0002	0.1396	1.0000

Question 5 (c)

```
. margins, at(age=26 female=0 bachelor=0) at(age=30 female=1 bachelor=1)

Adjusted predictions                            Number of obs     =      7,711
Model VCE    : Robust

Expression   : Linear prediction, predict()

1._at        : age             =          26
               female          =           0
               bachelor        =           0

2._at        : age             =          30
               female          =           1
               bachelor        =           1
```

	Margin	Delta-method Std. Err.	t	P>|t|	[95% Conf.	Interval]
_at						
1	14.57988	.1928156	75.62	0.000	14.20191	14.95785
2	21.33971	.1908257	111.83	0.000	20.96564	21.71378

For Bob:

$$\widehat{ahe} = -0.63 + 0.58 \times 26 - 3.66 \times 0 + 8.08 \times 0 = 14.45$$

For Jane:

$$\widehat{ahe} = -0.63 + 0.58 \times 30 - 3.66 \times 1 + 8.08 \times 1 = 21.19$$

(approximate numbers are fine)